Fisher information

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Outline

Regularity conditions

Let $\{f(\mathbf{x},\theta),\theta \in \Theta\}$ be a system of PDFs (PMFs). This system is regular if the following conditions are met:

1) Parametric space Θ is nonempty, open set.
2) A set $M = \{\mathbf{x} : f(\mathbf{x},\theta) > 0\}$ does not depend on the value of θ.
3) For each $\mathbf{x} \in M$ a finite partial derivative $f'(\mathbf{x},\theta) = \frac{\partial f(\mathbf{x},\theta)}{\partial \theta}$ exists.
4) For all $\theta \in \Theta$ holds $\int_M f'(\mathbf{x},\theta)d\mathbf{x} = 0$
5) Integral:

$$J_n(\theta) = \int_M \left[\frac{f'(\mathbf{x},\theta)}{f(\mathbf{x},\theta)}\right]^2 f(\mathbf{x},\theta)d\mathbf{x}$$

   is finite and positive

Fisher information

Definition

If the regularity conditions hold, integral

$$J_n(\theta) = \int_M \left[\frac{f'(\mathbf{x},\theta)}{f(\mathbf{x},\theta)}\right]^2 f(\mathbf{x},\theta) d\mathbf{x}$$

is called Fisher information.

Alternative forms of of Fisher information

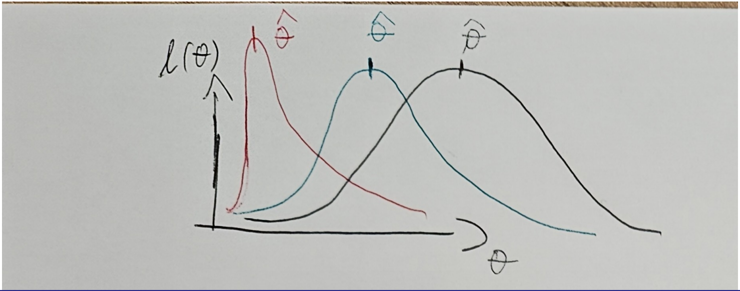
$$J_n(\theta) = \int_M \left[\frac{\partial l(\theta)}{\partial \theta}\right]^2 f(\mathbf{x},\theta) d\mathbf{x} = E\left[\frac{\partial l(\theta)}{\partial \theta}\right]^2$$

.

Intuitive meaning for Fisher information

Fisher information is a way to quantify a "quality of information" obtained by observing a random variable to estimate unknown parameter.

Under some additional assumptions, Fisher information can be used to compute variance of an estimator.



Theorem

If regularity conditions 3) and 4) can be extended to second derivatives, then Fisher information can be computed as follows:

$$J_n(\theta) = -E\left[\frac{\partial^2 l(\theta)}{\partial \theta^2}\right]$$

Theorem

Let $X_1, \ldots, X_n$ be IID random variables fulfilling regularity conditions, than Fisher information of the sample can be expressed as:

$$J_n(\theta) = nJ(\theta)$$

In order to generalize Fisher information for a parameter vector i is necessary to:

- expand regularity conditions for derivatives to all parameters
- take into account possible dependence between parameters. Meaning there will be a Fisher information for all pairs of i, j in a form

$$J_{i,j}(\theta) = \int_M \frac{\frac{\partial f(\mathbf{x},\theta)}{\partial \theta_i} \frac{\partial f(\mathbf{x},\theta)}{\partial \theta_j}}{f(\mathbf{x},\theta)^2} f(\mathbf{x},\theta) d\mathbf{x}$$

- hence instead of one value per parameter you get a Fisher information matrix, and positivity asumption on integral value needs to expanded to be positive definite matrix

Example

You are tasked with estimating a probability of creation of a defective part in your manufacturing process. However you are unsure about your sampling plan. You can gather the amount of controlled parts before a defect is found (geometric distribution), or you can record amount of defects in a set number of controlled parts (binomial distribution with known n). Try to decide on a sampling plan using Fisher information.

Python illustration

$$X_i \sim Ge(\pi)$$

$$p(x) = (1-\pi)^{x-1}(\pi) = L_i(\pi)$$

$$\ell(\pi) = (x-1)\cdot \ln(1-\pi) + \ln(\pi)$$

$$\frac{\partial \ell}{\partial \pi} = -\frac{x-1}{1-\pi} + \frac{1}{\pi}$$

$$\frac{\partial^2 \ell}{\partial \pi^2} = -\frac{x-1}{(1-\pi)^2} - \frac{1}{\pi^2}$$

$$J(\pi) = -E\left[\frac{\partial^2 \ell}{\partial \pi^2}\right] = -\sum_x \left(\frac{\partial^2 \ell}{\partial \pi^2}\right)\cdot p(x)$$

$$= E\left[\frac{x-1}{(1-\pi)^2} + \frac{1}{\pi^2}\right] = \frac{\overbrace{E(X)}^{=\frac{1}{\pi}}}{(1-\pi)^2} - \frac{1}{(1-\pi)^2} + \frac{1}{\pi^2} = \frac{1\pi - \pi^2 + (1-\pi)^2}{\pi^2(1-\pi)^2} = \frac{1}{\pi^2(1-\pi)}$$

$$Y \sim Bi(N, \pi)$$

$$p(y) = \binom{N}{y}(\pi)^y(1-\pi)^{N-y} \qquad J(\pi) = -E\left[\frac{\partial^2 \ell}{\partial \pi^2}\right]$$

$$\ell(\pi) = \ln\binom{N}{y} + y\cdot\ln(\pi) + (N-y)\cdot\ln(1-\pi)$$

$$\frac{\partial \ell}{\partial \pi} = \frac{y}{\pi} - \frac{N-y}{1-\pi}$$

$$\frac{\partial^2 \ell}{\partial \pi^2} = -\frac{y}{\pi^2} - \frac{N-y}{(1-\pi)^2}$$

$$J(\pi) = -E\left[\frac{\partial^2 \ell}{\partial \pi^2}\right] = \frac{\overbrace{E(Y)}^{=N\pi}}{\pi^2} + \frac{N - E(Y)}{(1-\pi)^2} = N\cdot\frac{\pi(1-\pi)^2 + (1-\pi)\pi^2}{\pi^2(1-\pi)^2} = \frac{N\cdot 1}{\pi(1-\pi)}$$

Rao-Crammér theorem (lower bound on the variance of an Unbiased estimator)

Theorem

Let $X_1, ..., X_n$ IID random variables with PDF (or PMF) satisfying all regularity conditions and $T = r(\mathbf{X})$ be an unbiased estimator of a differentiable parametric function $g(\theta)$ with finite variance. Then

$$Var_\theta(T) \geq \frac{[g'(\theta)]^2}{nJ(\theta).}$$

If the lower bound has been achieved by an estimator T^*, we call T^* efficient estimator.

Asymptotic distribution of an efficient estimator

Theorem

Assume the assumptions of Rao-Carmér theorem and let T be efficient estimator of a parametric function $g(\theta)$. Assume that $g'(\theta)$ is never equal to 0. Then the asymptotic distribution of

$$\frac{[nJ(\theta)]^{1/2}}{|g'(\theta)|}[T - g(\theta)]$$

is the standard normal distribution $N(0,1)$.

Asymptotic properties of an MLE estimator

Theorem

Assume that $\hat{\theta}$ is a consistent MLE estimator of a parameter θ obtained by solving $\frac{\partial L(\theta)}{\partial \theta} = 0$ or $\frac{\partial l(\theta)}{\partial \theta} = 0$. Assume further that regularity conditions hold, and that for all possible values of θ a third derivative of $L(\theta)$ exists (and satisfies some form of boundedness) and that $\int_M f''(x,\theta)dx = 0$. Then asymptotic distribution of

$$[nJ(\theta)]^{1/2}(\hat{\theta} - \theta)$$

is the standard normal distribution $N(0,1)$.

Python illustration

Under some strengthened consistency assumption, this can be generalized to iterative numerical solving of $\frac{\partial L(\theta)}{\partial \theta} = 0$ or $\frac{\partial l(\theta)}{\partial \theta} = 0$ by e.g. Newton-Raphson method.

Corollaries of knowing asymptotic distributions of an estimator:

- Construction of confidence intervals for parameters of "any" distribution
- Wald's test statistic $[nJ(\hat{\theta})](\hat{\theta} - \theta_0)^2 \sim \chi^2(1)$
- Likelihood ratio tests

Confidence interval

Definition

Let $X_1, \ldots, X_n$ be IID random variables, that depend on a parameter θ and $g(\theta)$ be a real valued parametric function. Let $A \leq B$ be two statistics that have a property

$$P(A \leq g(\theta) \leq B) \geq (1 - \alpha).$$

Then the random interval (A, B) is called $(1 - \alpha)$ confidence interval for $g(\theta)$.

Example: Derive a confidence interval for a parameter λ of exponential distribution.

$X_1, \dots, X_n$ ... i.i.d.

$X_i \sim Exp(\lambda)$ $\qquad \hat{\lambda} = \frac{1}{\bar{X}}$

$f(x) = \lambda e^{-\lambda x} \to \ell(\lambda) = \ln(\lambda) - \lambda x$

$\frac{\partial \ell}{\partial \lambda} = \frac{1}{\lambda} - x \qquad \frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{1}{\lambda^2}$

$U \sim N(0;1)$

$-u_{1-\alpha} = u_\alpha$

$J(\lambda) = -E\left[\frac{\partial^2 \ell}{\partial \lambda^2}\right] = \frac{1}{\lambda^2}$

$nJ(\lambda) = J_n(\lambda) \qquad \sqrt{nJ(\lambda)} \cdot (\hat{\lambda} - \lambda) \sim N(0;1)$

$$P\left(\lambda \in \left\langle \hat{\lambda} - u_{1-\alpha/2} \cdot \frac{1}{\sqrt{nJ(\lambda)}} ; \hat{\lambda} + u_{1-\alpha/2} \cdot \frac{1}{\sqrt{nJ(\lambda)}} \right\rangle\right) \doteq 1-\alpha$$

$\lambda \approx \hat{\lambda}$

$$\left\langle \hat{\lambda} - u_{1-\alpha/2} \cdot \frac{\hat{\lambda}}{\sqrt{n}} ; \hat{\lambda} + u_{1-\alpha/2} \cdot \frac{1\hat{\lambda}}{\sqrt{n}} \right\rangle$$

Likelihood ratio

Definition

Let θ_0 and θ_1 be possible values of a parameter θ of a distribution satisfying regularity conditions. Then likelihood ratio is defined:

$$\Lambda = \frac{L(\theta_0)}{L(\theta_1)}$$

.

Taking a natural logarithm of a likelihood ratio we get $ln(\Lambda) = l(\theta_0) - l(\theta_1)$.

Both forms can be used to compare which of the two parameter values fits the data better.

Likelihood ratio test

Definition

Under the same assumptions as for asymptotic distribution of MLE and given a hypothesized value of a parameter θ_0. The asymptotic distribution for the following statistic is $\chi^2(1)$.

$$LR(\theta_0) = 2[l(\hat{\theta}) - l(\theta_0)]$$

The likelihood ratio test holds under numerous generalizations and is extensively used in generalized linear models.

$$L(\lambda) = \prod_{i=1}^{x_i} \lambda e^{-\lambda x_i}$$

$$\ell(\lambda) = n \cdot \ln(\lambda) - \lambda \sum_{i=1}^{n} x_i \qquad J(\lambda) = \frac{1}{\lambda^2}$$

$H_0: \lambda = \lambda_0$

$H_1: \lambda \neq \lambda_1 \qquad \hat{\lambda} = \frac{1}{\bar{x}}$

$$LR = 2\left[n \cdot \ln\left(\frac{1}{\bar{x}}\right) - \frac{1}{\bar{x}} \cdot \sum x_i - \left(n \ln(\lambda_0) - \lambda_0 \sum x_i\right)\right]$$

$$LR = 2\left[n \cdot \ln\left(\frac{1}{\bar{x} \cdot \lambda_0}\right) - \sum_i (x_i)\left(\frac{1}{\bar{x}} - \lambda_0\right)\right] \overset{as}{\sim} \chi^2(1)$$

$$W = n \cdot J(\hat{\lambda})\left(\hat{\lambda} - \lambda_0\right)^2 = n \cdot \bar{x}^2 \left(\frac{1}{\bar{x}} - \lambda_0\right)^2 \overset{as}{\sim} \chi^2(1)$$